



Embedded Software for the Internet Of Things

TI DriverLib

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What is DriverLib?

- The Texas Instruments MSP432 Driver Library (DriverLib)
 - A set of fully functional APIs to configure, control, and manipulate the hardware peripherals of the MSP432 platform
- The DriverLib is a **software layer** to the programmer:
 - facilitates higher level of programming compared to direct register accesses
- **Not a layer of intelligence** that provides things automatically!

What is DriverLib?

Traditional Register Access

```
CSKEY = 0x695A;  
CSCTL1 |= SELM_1 | DIVM_2;  
CSKEY = 0
```



```
CS_initClockSignal(CS_MCLK, CS_VLOCLK_SELECT, CS_CLOCK_DIVIDER_32);
```

DriverLib Equivalent

DriverLib GPIO

```
#include "msp.h"
#include <ti/devices/msp432p4xx/driverlib/driverlib.h>

int main(void){
    volatile uint32_t i;

    // Stop watchdog timer
    WDT_A_holdTimer();
    // Set P1.0 to output direction
    GPIO_setAsOutputPin(GPIO_PORT_P1,GPIO_PIN0);

    while(1)    {
        // Toggle P1.0 output
        GPIO_toggleOutputOnPin(GPIO_PORT_P1,GPIO_PIN0);
        // Delay
        for(i=100000; i>0; i--);
    }
}
```

DriverLib GPIO

```
#include "msp.h"
#include <ti/devices/msp432p4xx/driverlib/driverlib.h>

int main(void){
    volatile uint32_t i;

    // Stop watchdog timer
    WDT_A_holdTimer();
    // Set P1.0 to output direction
    GPIO_setAsOutputPin(GPIO_PORT_P1,GPIO_PIN0);

    while(1)    {
        // Toggle P1.
        GPIO_toggleOut
        // Delay
        for(i=100000;

    }
}
```

void WDT_A_holdTimer (void)

Holds the Watchdog Timer.

This function stops the watchdog timer from running. This way no interrupt or PUC is asserted.

Returns

None

DriverLib GPIO

```
#include "msp.h"
#include <ti/devices/msp432p4xx/c...

int main(void){
    volatile uint32_t i;

    // Stop watchdog timer
    WDT_A_holdTimer();
    // Set P1.0 to output direction
    GPIO_setAsOutputPin(GPIO_PORT_...

    while(1)    {
        // Toggle P1.0 output
        GPIO_toggleOutputOnPin(GPI...
        // Delay
        for(i=100000; i>0; i--);
    }
}
```

```
void GPIO_setAsOutputPin ( uint_fast8_t selectedPort, uint_fast16_t
selectedPins )
```

This function configures the selected Pin as output pin.

This function selected pins on a selected port as output pins.

Parameters

selectedPort is the selected port. Valid values are:

- GPIO_PORT_P1
- GPIO_PORT_P2
- GPIO_PORT_P3
- GPIO_PORT_P4
- GPIO_PORT_P5
- GPIO_PORT_P6
- GPIO_PORT_P7
- GPIO_PORT_P8
- GPIO_PORT_P9
- GPIO_PORT_P10
- GPIO_PORT_P11
- GPIO_PORT_PJ

selectedPins is the specified pin in the selected port. Mask value is the logical OR of any of the following:

- GPIO_PIN0
- GPIO_PIN1
- GPIO_PIN2
- GPIO_PIN3
- GPIO_PIN4
- GPIO_PIN5
- GPIO_PIN6
- GPIO_PIN7
- GPIO_PIN8
- GPIO_PIN9
- GPIO_PIN10
- GPIO_PIN11
- GPIO_PIN12
- GPIO_PIN13
- GPIO_PIN14
- GPIO_PIN15

Modified bits of **PxDIR** register and bits of **PxSEL** register.

Returns

None

DriverLib GPIO

```
#include "msp.h"
#include <ti/devices/msp432p4xx/driverlib/driverlib.h>

int main(void){
    volatile uint32_t i;

    // Stop watchdog timer
    WDT_A_holdTimer();
    // Set P1.0 to output direction
    GPIO_setAsOutputPin(GPIO_PORT_P1,GPIO_PIN0);

    while(1)    {
        // Toggle P1.0 output
        GPIO_toggleOutputOnPin(GPIO_PORT_P1,GPIO_PIN0);
        // Delay
        for(i=100000; i>0; i--);
    }
}
```

Lab #1: Blinking LEDs

- Modify SingleLED file to blink multiple LEDs on your board.

DriverLib ISR

```
void port6_isr(void) {  
  
    uint32_t status = GPIO_getEnabledInterruptStatus(GPIO_PORT_P6);  
    GPIO_clearInterruptFlag(GPIO_PORT_P6, status);  
  
    if (status & GPIO_PIN7) {  
        stateChange = true;  
    }  
}
```

DriverLib ISR

```
void port6_isr(void) {  
  
    uint32_t status = GPIO_getEnabledInt  
    GPIO_clearInterruptFlag(GPIO_PORT_P6  
  
    if (status & GPIO_PIN7){  
        stateChange = true;  
    }  
}
```

uint_fast16_t GPIO_getEnabledInterruptStatus (uint_fast8_t selectedPort)

This function gets the interrupt status of the provided PIN and masks it with the interrupts that are actually enabled. This is useful for inside ISRs where the status of only the enabled interrupts needs to be checked.

Parameters

selectedPort is the selected port. Valid values are:

- GPIO_PORT_P1
- GPIO_PORT_P2
- GPIO_PORT_P3
- GPIO_PORT_P4
- GPIO_PORT_P5
- GPIO_PORT_P6
- GPIO_PORT_P7
- GPIO_PORT_P8
- GPIO_PORT_P9
- GPIO_PORT_P10
- GPIO_PORT_P11
- GPIO_PORT_PJ

Returns

Logical OR of any of the following:

- GPIO_PIN0
- GPIO_PIN1
- GPIO_PIN2
- GPIO_PIN3
- GPIO_PIN4
- GPIO_PIN5
- GPIO_PIN6
- GPIO_PIN7
- GPIO_PIN8
- GPIO_PIN9
- GPIO_PIN10
- GPIO_PIN11
- GPIO_PIN12
- GPIO_PIN13
- GPIO_PIN14
- GPIO_PIN15,
- PIN_ALL8,
- PIN_ALL16

indicating the interrupt status of the selected pins [Default: 0]

DriverLib ISR

```
void port6_isr(void) {  
  
    uint32_t status = GPIO_getEnabl  
    GPIO_clearInterruptFlag(GPIO_PO  
  
    if (status & GPIO_PIN7) {  
        stateChange = true;  
    }  
}
```

void GPIO_clearInterruptFlag (uint_fast8_t selectedPort, uint_fast16_t selectedPins)

This function clears the interrupt flag on the selected pin.

This function clears the interrupt flag on the selected pin.

Parameters

<i>selectedPort</i>	is the selected port. Valid values are: ■ GPIO_PORT_P1 ■ GPIO_PORT_P2 ■ GPIO_PORT_PA
<i>selectedPins</i>	is the specified pin in the selected port. Mask value is the logical OR of any of the following: ■ GPIO_PIN0 ■ GPIO_PIN1 ■ GPIO_PIN2 ■ GPIO_PIN3 ■ GPIO_PIN4 ■ GPIO_PIN5 ■ GPIO_PIN6 ■ GPIO_PIN7 ■ GPIO_PIN8 ■ GPIO_PIN9 ■ GPIO_PIN10 ■ GPIO_PIN11 ■ GPIO_PIN12 ■ GPIO_PIN13 ■ GPIO_PIN14 ■ GPIO_PIN15

Modified bits of PxIFG register.

Returns

None

Lab #2: GPIO ISR

- Capture button push events and blink different LEDs.

DriverLib Timers

```
/* Starting and enabling ACLK (32kHz) */
CS_setReferenceOscillatorFrequency(CS_REFO_128KHZ);
CS_initClockSignal(CS_ACLK, CS_REFOCLK_SELECT, CS_CLOCK_DIVIDER_4);
/* Configuring Continuous Mode */
Timer_A_configureContinuousMode(TIMER_A0_BASE, &continuousModeConfig);
/* Enabling interrupts and going to sleep */
Interrupt_enableSleepOnIsrExit();
Interrupt_enableInterrupt(INT_TA0_N);
/* Enabling MASTER interrupts */
Interrupt_enableMaster();
/* Starting the Timer_A0 in continuous mode */
Timer_A_startCounter(TIMER_A0_BASE, TIMER_A_CONTINUOUS_MODE);
```

DriverLib Timers

```
/* Starting and enabling ACLK (32kHz) */
CS_setReferenceOscillatorFrequency(CS_REFO_128KHZ);
CS_initClockSignal(CS_ACLK, CS_REFOCLK_SELECT, CS_CLOCK_DIVIDER_4);
/* Configuring Continuous Mode */
Timer_A_configureContinuousMode(TIMER_A0_BASE, &continuousModeConfig);
/* Enabling interrupts and going to sleep */
Interrupt_enableSleepOnIsrExit();
Interrupt_enableInterrupt(INT_TA0_N);
/* Enabling MASTER interrupts */
Interrupt_enableMaster();
/* Starting the Timer_A0 in continuous mode */
Timer_A_startCounter(TIMER_A0_BASE, TIMER_A_CONTINUOUS_MODE);
```

DriverLib Timers

```
void CS_setReferenceOscillatorFrequency ( uint8_t referenceFrequency )
```

Selects between the frequency of the internal REFO clock source

Parameters

<i>referenceFrequency</i>	selects between the valid frequencies: ■ CS_REFO_32KHZ, ■ CS_REFO_128KHZ,
---------------------------	---

Returns

NONE

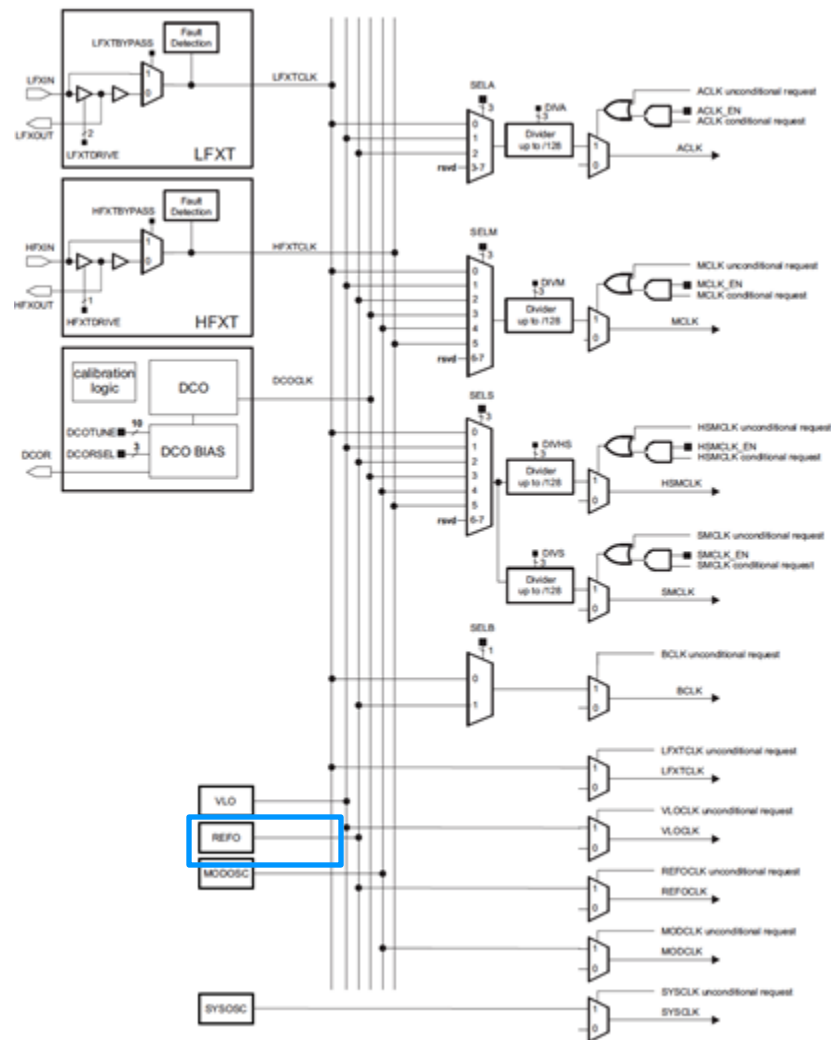


Figure 5-1. Clock System Block Diagram

DriverLib Timers

```
void CS_initClockSignal ( uint32_t selectedClockSignal, uint32_t clockSource,
uint32_t clockSourceDivider )
```

This function initializes each of the clock signals. The user must ensure that this function is called for each clock signal. If not, the default state is assumed for the particular clock signal. Refer to DriverLib documentation for CS module or Device Family User's Guide for details of default clock signal states.

Note that this function is blocking and will wait on the appropriate bit to be set in the CSSTAT READY register to be set before setting the clock source.

Also note that when HSMCLK and SMCLK share the same clock signal. If you change the clock signal for HSMCLK, the clock signal for SMCLK will change also (and vice-versa).

HFXTCLK is not available for BCLK or ACLK.

Parameters

selectedClockSignal	Clock signal to initialize. ■ CS_ACLK, ■ CS_MCLK, ■ CS_HSMCLK ■ CS_SMCLK ■ CS_BCLK [clockSourceDivider is ignored for this parameter]
clockSource	Clock source for the selectedClockSignal signal. ■ CS_LFXTCLK_SELECT, ■ CS_HFXTCLK_SELECT, ■ CS_VLOCLK_SELECT, [Not available for BCLK] ■ CS_DCOCLK_SELECT, [Not available for ACLK, BCLK] ■ CS_REFOCLK_SELECT, ■ CS_MODOSC_SELECT [Not available for ACLK, BCLK]
clockSourceDivider	selected the clock divider to calculate clock signal from clock source. This parameter is ignored when setting BCLK. Valid values are: ■ CS_CLOCK_DIVIDER_1, ■ CS_CLOCK_DIVIDER_2, ■ CS_CLOCK_DIVIDER_4, ■ CS_CLOCK_DIVIDER_8, ■ CS_CLOCK_DIVIDER_16, ■ CS_CLOCK_DIVIDER_32, ■ CS_CLOCK_DIVIDER_64, ■ CS_CLOCK_DIVIDER_128

Returns
NONE

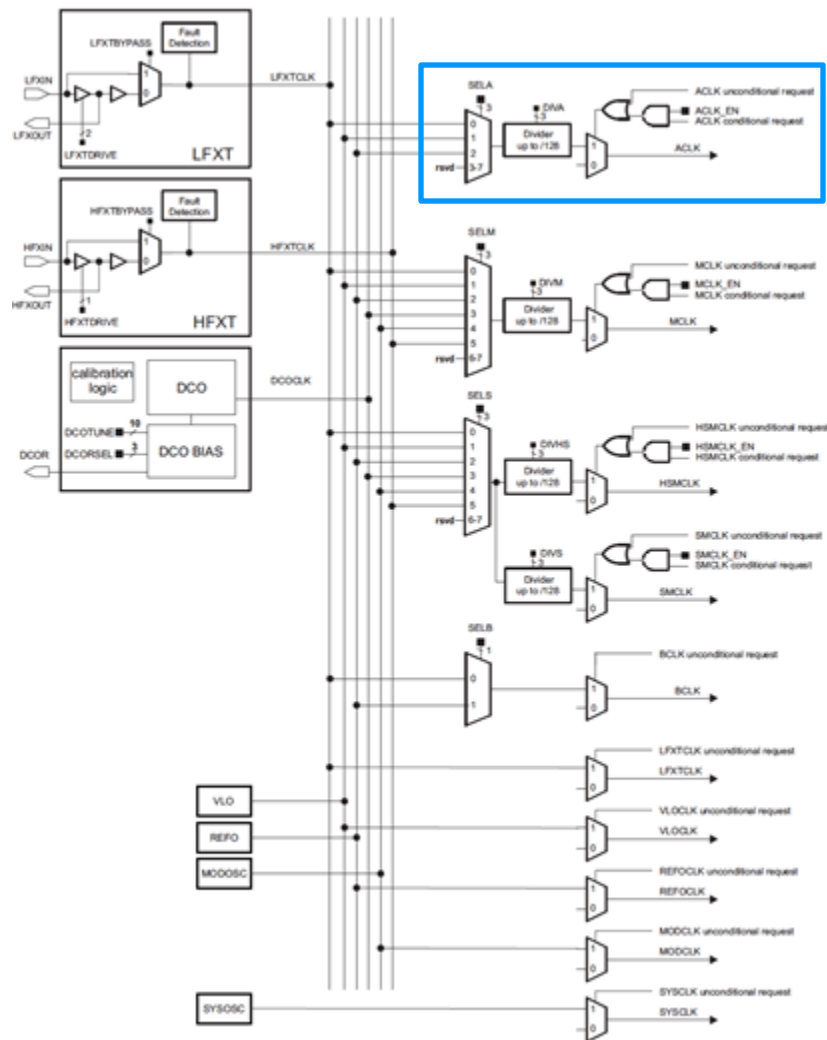


Figure 5-1. Clock System Block Diagram

DriverLib Timers

```
void Timer_A_configureContinuousMode ( uint32_t timer, const  
Timer_A_ContinuousModeConfig * config )
```

Configures Timer_A in continuous mode.

Parameters

<i>timer</i>	is the instance of the Timer_A module. Valid parameters vary from part to part, but can include: ■ TIMER_A0_BASE ■ TIMER_A1_BASE ■ TIMER_A2_BASE ■ TIMER_A3_BASE
<i>config</i>	Configuration structure for Timer_A continuous mode

Configuration options for Timer_A_ContinuousModeConfig structure.

```
/* Starting and enabling ACLK (32k  
CS_setReferenceOscillatorFrequency  
CS_initClockSignal(CS_ACLK, CS_REF
```

```
/* Configuring Continuous Mode */
```

```
Timer_A_configureContinuousMode(TIMER_A0_BASE, &continuousModeConfig);
```

```
/* Enabling interrupts and going to sleep */
```

```
Interrupt_enableSleepOnIsrExit();
```

```
Interrupt_enableInterrupt(INT_TA0_N);
```

```
/* Enabling MASTER
```

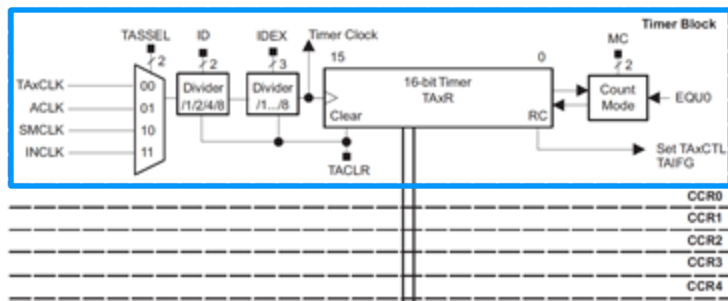
```
Interrupt_enableM
```

```
/* Starting the T
```

```
Timer_A_startCoun
```

```
const Timer_A_ContinuousModeConfig continuousModeConfig =  
{  
    TIMER_A_CLOCKSOURCE_ACLK,           // ACLK Clock Source  
    TIMER_A_CLOCKSOURCE_DIVIDER_1,      // ACLK/1 = 32.768khz  
    TIMER_A_TAIE_INTERRUPT_ENABLE,      // Enable Overflow ISR  
    TIMER_A_DO_CLEAR                     // Clear Counter  
};
```

DriverLib Timers



```
const Timer_A_ContinuousModeConfig continuousModeConfig =
{
    TIMER_A_CLOCKSOURCE_ACLK,           // ACLK Clock Source
    TIMER_A_CLOCKSOURCE_DIVIDER_1,      // ACLK/1 = 32.768khz
    TIMER_A_TAIE_INTERRUPT_ENABLE,      // Enable Overflow ISR
    TIMER_A_DO_CLEAR                    // Clear Counter
};
```

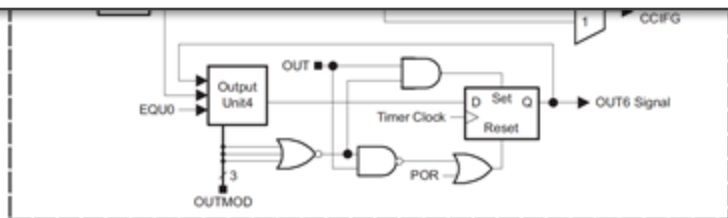


Figure 17-1. Timer_A Block Diagram

Parameters	
<i>clockSource</i>	selects Clock source. Valid values are ■ <code>TIMER_A_CLOCKSOURCE_EXTERNAL_TXCLK</code> [Default value] ■ <code>TIMER_A_CLOCKSOURCE_ACLK</code> ■ <code>TIMER_A_CLOCKSOURCE_SMCLK</code> ■ <code>TIMER_A_CLOCKSOURCE_INVERTED_EXTERNAL_TXCLK</code>
<i>timerInterruptEnable_TAIE</i>	is the divider for Clock source. Valid values are: ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1</code> [Default value] ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_8</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_16</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_32</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_64</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_128</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_256</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_512</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1024</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2048</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4096</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_8192</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_16384</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_32768</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_65536</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_131072</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_262144</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_524288</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1048576</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2097152</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4194304</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_8388608</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_16777216</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_33554432</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_67108864</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_134217728</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_268435456</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_536870912</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1073741824</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2147483648</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4294967296</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_8589934592</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_17179869184</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_34359738368</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_68719476736</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_137438953472</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_274877906944</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_549755813888</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1099511627776</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2199023255552</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4398046511104</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_8796093022208</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_17592186044416</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_35184372088832</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_70368744177664</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_140737488355328</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_281474976710656</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_562949953421312</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1125899906842624</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2251799813685248</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4503599627370496</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_9007199254740992</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_18014398509481984</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_36028797018963968</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_72057594037927936</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_144115188075855872</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_288230376151711744</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_576460752303423488</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1152921504606846976</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2305843009213693952</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4611686018427387904</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_9223372036854775808</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_18446744073709551616</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_36893488147419103232</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_73786976294838206464</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_147573952589676412928</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_295147905179352825856</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_590295810358705651712</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1180591620717411303424</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2361183241434822606848</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4722366482869645213696</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_9444732965739290427392</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_18889465931478580854784</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_37778931862957161709568</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_75557863725914323419136</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_151115727451828646838272</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_302231454903657293676544</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_604462909807314587353088</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1208925819614629174706176</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2417851639229258349412352</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4835703278458516698824704</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_9671406556917033397649408</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_19342813113834066795298816</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_38685626227668133590597632</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_77371252455336267181195264</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_154742504910672534362390528</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_309485009821345068724781056</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_618970019642690137449562112</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1237940039285380274899124224</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2475880078570760549798248448</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_4951760157141521099596496896</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_9903520314283042199192993792</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_19807040628566084398385987584</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_39614081257132168796771975168</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_79228162514264337593543950336</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_158456325028528675187087900672</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_316912650057057350374175801344</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_633825300114114700748351602688</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1267650600228229401496703205376</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2535301200456458802993406410752</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5070602400912917605986812821504</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_10141204801825835211973625643008</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_20282409603651670423947251286016</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_40564819207303340847894502572032</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_81129638414606681695789005144064</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_162259276829213363391578010288128</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_324518553658426726783156020576256</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_649037107316853453566312041152512</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1298074214633706907132624082305024</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2596148429267413814265248164610048</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5192296858534827628530496329220096</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_10384593717069655257060992658440192</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_20769187434139310514121985316880384</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_41538374868278621028243970633760768</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_83076749736557242056487941267521536</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_166153499473114484112975882535043072</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_332306998946228968225951765070086144</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_664613997892457936451903530140172288</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1329227995784915872903807060280344576</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2658455991569831745807614120560689152</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5316911983139663491615228241121378304</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_10633823966279326983230456482242756608</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_21267647932558653966460912964485513216</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_42535295865117307932921825928971026432</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_85070591730234615865843651857942052864</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_170141183460469231731687303715884105728</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_340282366920938463463374607431768211456</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_680564733841876926926749214863536422912</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1361129467683753853853498429727072845824</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2722258935367507707706996859454145691648</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5444517870735015415413993718908291383296</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_10889035741470030830827987437816582766592</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_21778071482940061661655974875633165533184</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_43556142965880123323311949751266331066368</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_87112285931760246646623899502532662132736</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_174224571863520493293247799005065324265472</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_348449143727040986586495598010130648530944</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_696898287454081973172991196020261297061888</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1393796574908163946345982392040522594123776</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2787593149816327892691964784081045188247552</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5575186299632655785383929568162090376495104</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_11150372599265311570767859136324180752990208</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_22300745198530623141535718272648361505980416</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_44601490397061246283071436545296723011960832</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_89202980794122492566142873090593446023921664</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_178405961588244985132285746181186892047843328</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_356811923176489970264571492362373784095686656</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_713623846352979940529142984724747568191373312</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1427247692705959881058285969449495136382746624</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2854495385411919762116571938898990272765493248</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5708990770823839524233143877797980545530986496</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_11417981541647679048466287755595961091061972992</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_22835963083295358096932575511191922182123945984</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_45671926166590716193865151022383844364247891968</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_91343852333181432387730302044767688728495783936</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_182687704666362864775460604089535377456991567872</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_365375409332725729550921208179070754913983135744</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_730750818665451459101842416358141509827966271488</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1461501637330902918203684832716283019655932542976</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2923003274661805836407369665432566039311865085952</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5846006549323611672814739330865132078623730171904</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_11692013098647223345629478661730264157247460343808</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_23384026197294446691258957323460528314494920687616</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_46768052394588893382517914646921056628989841375232</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_93536104789177786765035829293842113257979682750464</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_187072209578355573530071658587684226515959365500928</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_374144419156711147060143317175368453031918731001856</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_748288838313422294120286634350736906063837462003712</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1496577676626844588240573268701473812127674924007424</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_2993155353253689176481146537402947624255349848014848</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_5986310706507378352962293074805895248510699696029696</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_11972621413014756705924586149611790497021399392059392</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_23945242826029513411849172299223580994042798784118784</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_47890485652059026823698344598447161988085597568237568</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_95780971304118053647396689196894323976171195136475136</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_191561942608236107294793378393788647952342390272950272</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_383123885216472214589586756787577295904684780545900544</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_766247770432944429179173513575154591809369561091801088</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1532495540865888858358347027150309183618739122183602176</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_3064991081731777716716694054300618367237478244367204352</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_6129982163463555433433388108601236734474956488734408704</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_12259964326927110866866776217202473468949912977468817408</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_24519928653854221733733552434404946937899825954937634816</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_49039857307708443467467104868809893875799651909875269632</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_98079714615416886934934209737619787751599303819750539264</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_196159429230833773869868419475239575503198607639501078528</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_392318858461667547739736838950479151006397215279002157056</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_784637716923335095479473677900958302012794430558004314112</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1569275433846670190958947355801916604025588861116008628224</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_3138550867693340381917894711603833208051177722232017256448</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_6277101735386680763835789423207666416102355444464034512896</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_12554203470773361527671578846415332832204710888928069025792</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_25108406941546723055343157692830665664409421777856138051584</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_50216813883093446110686315385661331328818843555712276103168</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_100433627766186892221372630771322662657637687111424552206336</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_200867255532373784442745261542645325315275374222849104412672</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_401734511064747568885490523085290650630550748445698208825344</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_803469022129495137770981046170581301261101496891396417650688</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1606938044258990275541962092341162602522202993782792835301376</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_3213876088517980551083924184682325205044405987565585670602752</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_6427752177035961102167848369364650410088811975131171341205504</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_12855504354071922204335696738729300820177623950262342682411008</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_25711008708143844408671393477458601640355247900524685364822016</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_51422017416287688817342786954917203280710495801049370729644032</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_102844034832575377634685573909834406561420991602098741459288064</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_205688069665150755269371147819668813122841983204197482918576128</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_411376139330301510538742295639337626245683966408394965837152256</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_822752278660603021077484591278675252491367932816789931674304512</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_1645504557321206042154969182557350504982735865633579863348609024</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_3291009114642412084309938365114701009965471731267159726697218048</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_6582018229284824168619876730229402019930943462534319453394436096</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_13164036458569648337239753460458804039861886925068638906788872192</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_26328072917139296674479506920917608079723773850137277813577744384</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_52656145834278593348959013841835216159447547700274555627155488768</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_105312291668557186697918027683670432318895095400549111254310977536</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_210624583337114373395836055367340864637790190801098222508621955072</code> ■ <code>TIMER_A_CLOCKSOURCE_DIVIDER_421249166674228746791672110734681729275580381602196445017243910144</code>

DriverLib Timers

```
/* Starting and enabling ACLK (32kHz) */
CS_setReferenceOscillatorFrequency(CS_REFCLK_S, CS_32K);
CS_initClockSignal(CS_ACLK, CS_REFCLK_S, CS_ACLK);
/* Configuring Continuous Mode */
Timer_A_configureContinuousMode(TIMER_A0, TA_MODE_CONTINUOUS);
/* Enabling interrupts and going to sleep */
Interrupt_enableSleepOnIsrExit();
Interrupt_enableInterrupt(INT_TA0_N);
/* Enabling MASTER interrupts */
Interrupt_enableMaster();
/* Starting the Timer_A0 in continuous mode */
Timer_A_startCounter(TIMER_A0_BASE, TIME_0);
```

void Interrupt_enableInterrupt (uint32_t interruptNumber)

Enables an interrupt.

Parameters

interruptNumber specifies the interrupt to be enabled.

The specified interrupt is enabled in the interrupt controller. Other enables for the interrupt (such as at the peripheral level) are unaffected by this function.

Valid values will vary from part to part, so it is important to check the device specific datasheet, however for MSP432 101 the following values can be provided:

- FAULT_NMI
- FAULT_HARD
- FAULT_MPU
- FAULT_BUS
- FAULT_USAGE
- FAULT_SVCALL
- FAULT_DEBUG
- FAULT_PENDSV
- FAULT_SYSTICK
- INT_PSS
- INT_CS
- INT_PCM
- INT_WDT_A
- INT_FPU
- INT_FLCTL
- INT_COMP0
- INT_COMP1
- INT_TA0_0
- INT_TA0_N
- INT_TA1_0
- INT_TA1_N
- INT_TA2_0
- INT_TA2_N
- INT_TA3_0
- INT_TA3_N
- INT_EUSCIA0
- INT_EUSCIA1
- INT_EUSCIA2

DriverLib Timers

```
bool Interrupt_enableMaster ( void )
```

Enables the processor interrupt.

This function allows the processor to respond to interrupts. This function does not affect the set of interrupts enabled in the interrupt controller; it just gates the single interrupt from the controller to the processor.

Returns

Returns **true** if interrupts were disabled when the function was called or **false** if they were initially enabled.

```
/* Starting and enabling
```

```
CS_setReferenceOscillator
```

```
CS_initClockSignal(CS_ACI
```

```
/* Configuring Continuous
```

```
Timer_A_configureContinuousMode(TIMER_A0_BASE, &continuousModeConfig);
```

```
/* Enabling interrupts and going to sleep */
```

```
Interrupt_enableSleepOnIsrExit();
```

```
Interrupt_enableInterrupt(INT_TA0_N);
```

```
/* Enabling MASTER interrupts */
```

```
Interrupt_enableMaster();
```

```
/* Starting the Timer_A0 in continuous mode */
```

```
Timer_A_startCounter(TIMER_A0_BASE, TIMER_A_CONTINUOUS_MODE);
```

DriverLib Timers

```
/* Starting and enabling ACLK (321
CS_setReferenceOscillatorFrequency
CS_initClockSignal(CS_ACLK, CS_REI
/* Configuring Continuous Mode */
Timer_A_configureContinuousMode(T
/* Enabling interrupts and going
Interrupt_enableSleepOnIsrExit();
Interrupt_enableInterrupt(INT_TAO
/* Enabling MASTER interrupts */
Interrupt_enableMaster();
/* Starting the Timer_A0 in continuous mode */
Timer_A_startCounter(TIMER_A0_BASE, TIMER_A_CONTINUOUS_MODE);
```

```
void Timer_A_startCounter ( uint32_t timer, uint_fast16_t timerMode )
```

Starts Timer_A counter

Parameters

<i>timer</i>	is the instance of the Timer_A module. Valid parameters vary from part to part, but can include: ■ TIMER_A0_BASE ■ TIMER_A1_BASE ■ TIMER_A2_BASE ■ TIMER_A3_BASE
<i>timerMode</i>	selects Clock source. Valid values are ■ TIMER_A_CONTINUOUS_MODE [Default value] ■ TIMER_A_UPDOWN_MODE ■ TIMER_A_UP_MODE

Note

This function assumes that the timer has been previously configured using Timer_A_configureContinuousMode, Timer_A_configureUpMode or Timer_A_configureUpDownMode.

- Two interrupt vectors are associated with the 16-bit Timer_A module:
 - TAxCCR0 interrupt vector for TAxCCR0 CCIFG
 - TAxIV interrupt vector for all other CCIFG flags and TAIFG



Figure 17-1. Timer_A Block Diagram

DriverLib Timers

- TAxCCR0 interrupt vector for TAxCCR0 CCIFG
 - **TA0_0_IRQHandler**
- TAxIV interrupt vector for all other CCIFG flags and TAIFG
 - **TA0_N_IRQHandler**

```
void TA0_N_IRQHandler(void)
{
    Timer_A_clearInterruptFlag(TIMER_A0_BASE);
    GPIO_toggleOutputOnPin(GPIO_PORT_P1, GPIO_PIN0);
}
```

DriverLib Timers

- TAxCCR0 interrupt vector for T
 - TA0_0_IRQHandler
- TAxIV interrupt vector for all o and TAIFG
 - TA0_N_IRQHandler

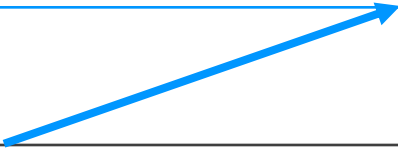
```
void Timer_A_clearInterruptFlag ( uint32_t timer )
```

Clears the Timer TAIFG interrupt flag

Parameters

<i>timer</i>	is the instance of the Timer_A module. Valid parameters vary from part to part, but can include: ■ TIMER_A0_BASE ■ TIMER_A1_BASE ■ TIMER_A2_BASE ■ TIMER_A3_BASE
--------------	--

Returns
None



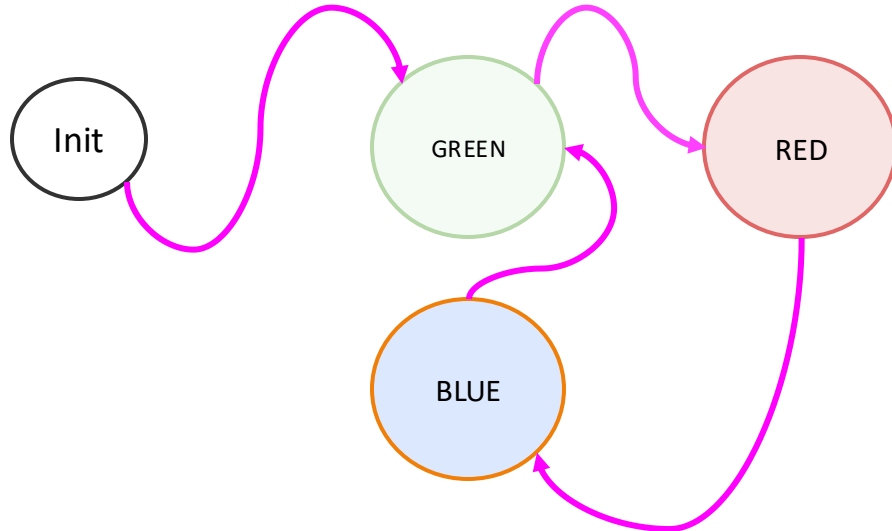
```
void TA0_N_IRQHandler(void)
{
    Timer_A_clearInterruptFlag(TIMER_A0_BASE);
    GPIO_toggleOutputOnPin(GPIO_PORT_P1, GPIO_PIN0);
}
```

Lab #3: Timers

- Compile and run up-mode timer on Moodle.
- Create a continuous-mode timer!

Lab #4: FSM

- Implement the following FSM with with **DriverLib** timers.
 - Each state transition will take place every 1 second.



Buttons and LEDs

